

Second part: Open book

Total score: 10 pts.

1. Consider a telecommunication network represented by a directed graph $G = (N, A)$. Root node r has to broadcast a message to every other node in N . At any one unit interval of time t a node can either receive or send a message to ONLY ONE of its neighboring nodes. Sending (or receiving) the message along an arc $(i, j) \in A$ takes 1 unit of time. A node can send the message only after it has received it. Find the routing operations during the time that minimize the time for broadcasting the message to all the nodes. Write a suitable linear optimization model. Explain the meaning of variables and constraints.
2. Consider a directed cycle defined on a node set N and such that $|N| = n$. The distance between any two consecutive nodes in the cycle i and $i + 1$ is given by d_{ii+1} and d_{n1} from n to 1. The distance between two non adjacent nodes i and j is given by the sum of the consecutive arcs in the cycle from i to j . Consider now the problem of partitioning the cycle in the most uniform way, that is selecting a subset of nodes $k < n$ such that the maximum distance between consecutive selected nodes is minimized. Provide a greedy algorithm.